

Risk Register

The following table is used to evaluate the level of the risks that may happen during using or testing the board. Rating for the priority level considers both the result caused by the risk and the probability of the event happening. The Risk Matrix shows the greater the priority level is, the more severe the risk is.

Risk Matrix

P R O B A B I L I T Y	5	5	10	15	20	25
	4	4	8	12	16	20
	3	3	6	9	12	15
	2	2	4	6	8	10
	1	1	2	3	4	5
		1	2	3	4	5
	IMPACT					

Risk description (Information about the situations)	Impact description (The result caused by the risk)	Impact level Rate 1(LOW) to 5(HIGH)	Probability level Rate 1(LOW) to 5(HIGH)	Priority level (Impact level x Probability level)	Responsible person	Response plan How to solve the problem or reduce the risk
Supply the input voltage without setting a sufficiently low current limit	May destroy the components in the circuit if there is any short circuit fault.	5	2	10	Arno Liu	Cut the power supply immediately and check each node and components.
Connect the directional components in an incorrect way	The DPS mayl not work as expected and may destroy certain components	5	3	15	Joshua Liao	Check the direction of each component in the whole circuit, especially components such as diodes and electrolytic capacitors.
Connect	The output	3	2	6	Chenyi Lu	Desolder the

components with wrong value in the circuit	voltages would be different from the expected values.					component, then solder the component with the correct value.
Supply a voltage below the minimum input voltage	The output will not give a sufficient voltage	3	2	6	Siyu Liu	Check the input voltage setting or battery voltage.
Supply a voltage above the maximum input voltage	The board may be destroyed	5	2	10	Lyuquan Gao	Test each node to see if the DPS is working correctly or not, replace damaged components. Otherwise the board is completely unusable.
Some inevitable changes when building a circuit board	change the efficiency of the circuit	1	4	4	Arno Liu	Change to another PCB board or take no action
The circuit was designed to have two independent and parallel output circuit	If 1 output stops working, the other one will not be affected	1	1	1	Arno Liu	Only need to check the faulty output circuit.
Did not obey the design rule when designing PCB layout	The PCB will not be correctly manufactured	4	3	12	Joshua Liao	Check the layout design and remanufacture the PCB.
Take too long time to place the components on the soldering paste	The components may not be fixed on the PCB causing poor connection or falling apart	4	2	8	Siyu Liu	Check the connectivity between components and the board. If there are any unstable components, re-solder them.